

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 3345	Course Title: Database Systems Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:1 T:0 P:2)	Periods per semester: 45

Course Objectives:

Students will be able to practice database problems using MySQL & MongoDB database.

Course Prerequisites:

Topic	Course Name	Semester
Knowledge of basics of Computer	Introduction to IT systems	1
Basic Knowledge of IT skills and basic problem solving	Problem solving and Programming	2

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the use of DDL and DML commands in MySQL database	8	Applying
CO2	Develop queries using aggregate functions, join operations, nested queries and views	9	Applying
CO3	Develop programs using functions, procedures, and trigger	12	Applying
CO4	Implement basic database operations in MongoDB.	13	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate the use of DDL and DML commands in MySQL database		
M1.01	Demonstrate DDL commands-CREATE, ALTER, DROP	1L +3P	Applying
M1.02	Apply integrity constraints - PRIMARY KEY, NOT NULL, DEFAULT, CHECK, UNIQUE, FOREIGN KEY		Applying
M1.03	Experiment DML commands - INSERT, SELECT, UPDATE and DELETE to manipulate databases	1L +3P	Applying
M1.04	Experiment SELECT statement using WHERE, DISTINCT, BETWEEN, ORDER BY, IN, LIKE for data retrieval from the databases		Applying
CO2	Develop queries using aggregate functions, join operations, nested queries and views		
M2.01	Implement aggregate functions - SUM, AVG, COUNT, MIN, MAX - GROUP BY and HAVING clause	2L +3P	Applying
M2.02	Construct nested queries		Applying
M2.03	Demonstrate queries for inner and outer joins.	1L +3P	Applying

M2.04	Illustrate various views of the table		Applying
	Lab Exam – I	1.5	
CO3	Develop programs using functions, procedures, and trigger		
M3.01	Develop programs using Stored Procedures	1L +5P	Applying
M3.02	Develop programs using Stored Functions		Applying
M3.03	Develop programs using Cursor	1L +5P	Applying
M3.04	Illustrate programs using triggers	1L +1P	Applying
CO4	Implement basic database operations in MongoDB		
M4.01	Install and configure MongoDB.	1P	Applying
M4.02	Implement CRUD operations in MongoDB.	1L +5P	Applying
	Open Ended Project	6	Applying
	Lab Exam – II	1.5	

****** Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation). Students can do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on an open -ended experiment of their choice.

Example: Develop databases for

- Employee payroll
- Customer billing system.
- Library management system

Text / Reference

T/R	Book Title/Author
T1	PHP and MySQL web development -Fifth edition-Luke Welling, Laura Thomson
T2	Sam's teach6 Yourself NoSQL with6 Mong5oDB- Brad Dayley

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/mysql/index.htm
2	https://www.tutorialspoint.com/mongodb/index.htm